

FUNDAMENTALS OF COMPUTER NETWORKS

CHAPTER 1 : Introduction to Networks

1.1 Network Components and Topologies

1.1.1 Network Components

Host / End Device : Any computer connected to a network

Example : Laptop, PC, Smartphone, Server

Explanation : A host is any device that can send or receive data on a network

Server and Client

Role	Explanation	Function	Example
Server	A computer that provides services or resources	Stores and manages data	Gmail server stores emails
Client	A computer that requests services or resources	Sends requests to servers	Your laptop checking email

Types of Servers

Server Type	Explanation	Client Software Used	Example
Email Server	Runs email server software to send, receive, and store emails	Outlook, Gmail App	Sending/receiving email
Web Server	Runs web server software to host websites	Web Browser (Chrome, Edge)	Viewing web pages
File Server	Stores company and user files for sharing	File Explorer, Drive	Downloading company files

Peer-to-Peer Network

A network where each device can act as both client and server (No dedicated central server)		
Device Role	Function	Example
Client (Computer A)	Requests data	Computer A requests file
Server (Computer B)	Provides data	Computer B sends file
Same Device	Can be both client & server	File sharing between PCs
Advantages and Disadvantages		
Advantages		Disadvantages
Easy to set up		No centralized administration 没有中央行政机构
Less complex		Not secure
Lower cost		Not scalable 不可扩展
Good for small networks		Performance may slow down

Peer-to-Peer Network – Usage / Suitable Tasks

Task Type	Example
File transfer	Sharing folders between PCs
Printer sharing	One printer used by many PCs
Home network	Small office / home setup

End Devices

A device where data starts or ends (Source or destination of data)		
Example: Computer / PC, Smartphone, Printer		
Scenario (Real-Life Example)	Source End Device	Destination End Device
Send email	Your laptop	Receiver's PC
Print document	Your computer	Network printer
Watch YouTube	Your phone	YouTube server (end system)

Intermediary Network Devices

A device that connects end devices together (Controls and manages data flow)		
Example: Switch, Router, Wireless Access Point, Firewall		
Role Intermediary Network Devices		
Role	Explanation	Example
Regenerate & retransmit signals	Strengthens weak data signals	Long cable transmission
Maintain pathway information	Knows best path for data	Router routing table
Error notification	Alerts devices about failures	Packet loss warning

Network Media

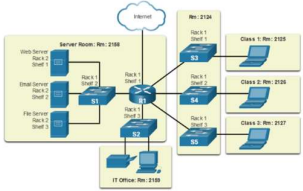
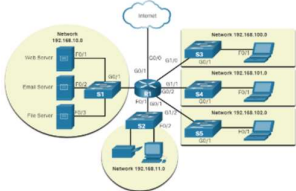
The channel that carries data from source to destination (Physical or wireless path)				
Media Types				
Media Type		Explanation	Signal Type	Example
Metal Wires (Copper Cable)		Uses electrical signals	Electrical impulses	Ethernet cable
Fiber-Optic Cable 光纤电缆		Uses light through glass/plastic fibers	Light pulses	High-speed internet
Wireless Transmission		Uses electromagnetic waves	Radio frequencies	Wi-Fi, Bluetooth
Comparison of Media Types				
Media	Speed	Cost	Interference	Common Use
Copper	Medium	Low	High	LAN cables
Fiber-Optic	Very High	Expensive	Very Low	Backbone internet
Wireless	Varies	Medium	Medium	Wi-Fi networks
Real-Life Examples				
Scenario		Media Used		
Home internet via LAN cable		Copper wire		
Submarine internet cable		Fiber optic		
Connecting phone to Wi-Fi		Wireless		

1.1.2 Network Representations and Topologies

Network Representations

Network Diagram (Topology Diagram) is a visual drawing that shows devices and connections in a network by using uses symbols/icons			
Purpose of Network Diagrams			
Purpose		Description	
Visualization		See network structure clearly	
Troubleshooting		Identify connection problems	
Planning		Design new networks	
Documentation		Record network layout	
Key Terms in Network Representation			
Term	Explanation	Function	Example
NIC (LAN Adapter)	Hardware that connects a device to a network	Provides physical network connection	Ethernet card in PC 
Physical Port	A socket/connector on a device	Where network media connects	Ethernet port on router
Interface	A specialized port on networking devices	Connects different networks	Router interfaces

Topology Diagrams

A diagram that shows the structure of a network (Physical or logical view)		
Physical Topology: Shows actual physical layout (Device location, cable installation), Focus: Hardware & cables		
Logical Topology: Shows how data flows in the network (Devices, ports, IP addressing), Focus: Data communication		
Physical Topology Diagram 物理拓扑		
Aspect	Details	Example
Device location	Where devices are placed	Office floor layout
Cable installation	How cables are wired	Ethernet cable paths
Hardware view	Physical connections	Switch to PC cabling
解：显示网络设备与网线在现实中的实际位置与连接方式。		
Logical Topology Diagram 逻辑拓扑		
Aspect	Details	Example
Devices	Network nodes	PCs, routers
Ports / Interfaces	Connection points	Router interfaces
Addressing scheme	IP addresses, VLANs	192.168.1.0 network
解：显示网络中数据传输的路径、设备连接关系与 IP 位址规划。		
Physical Topology Diagram 		Logical Topology Diagram 

1.2 Common Types of Networks

Network Sizes

Classification of networks based on scale and coverage (From small home to global networks)			
Network Type	Description	Scale	Example
Small Home Network	Connects a few devices at home to each other and the Internet	Very Small	Home Wi-Fi
Small Office / Home Office (SOHO)	Connects home or small office to corporate network	Small	Remote work setup
Medium to Large Network	Multiple locations with many interconnected computers	Large	Company branches
World Wide Network	Connects computers globally	Very Large	Internet

LANs and WANs

Aspect	Description	LAN Example	WAN Example
Area Covered	Geographical size of the network	Office building	Multiple countries
Number of Users	Total connected users/devices	Company staff in one building	Global company employees
Services Available	Types of services provided	File sharing, printers, local servers	Cloud services, global databases
Area of Responsibility	Who manages the network	Internal IT department	ISPs / Telecom providers
LANs and WANs Comparison			
Feature	LAN (Local Area Network)	WAN (Wide Area Network)	
Definition	Network covering a small geographical area	Network covering a wide geographical area	
Coverage Area	Limited (room, building, campus)	Very large (cities, countries, global)	
Connection Scope	Connects end devices directly	Connects multiple LANs together	
Administration	Managed by single organization/individual	Managed by service providers (ISPs)	
Bandwidth Speed	High-speed connections	Slower compared to LAN	
Ownership	Private ownership	Shared / public infrastructure	
Cost	Lower setup & maintenance cost	Higher cost due to large infrastructure	
Examples	School lab, office network, home Wi-Fi	Internet, bank network, multinational company network	
Technologies Used	Ethernet, Wi-Fi	MPLS, leased lines, satellite, fibre backbone	
Purpose	Internal communication & resource sharing	Long-distance communication	

The Internet

Aspect	Description	Details / Examples
Definition	A worldwide collection of interconnected networks 全球互联网络集合	Combines LANs and WANs globally
Network Connection	LANs are linked together using WANs	Office LAN → ISP WAN → Internet
Transmission Media	Technologies used to carry data 用于传输数据的技术	Copper wires, Fiber optic cables, Wireless signals
Ownership	Not owned by any single person or organization	Public global infrastructure
Governance	Managed through international organizations	Standards & coordination bodies
The following groups were developed to help maintain structure on the internet : - IETF (Develops internet protocols (e.g., TCP/IP)) - ICANN (Manages domain names & IP addresses) - IAB (Oversees internet design & standards)		

Intranet and Extranet

Feature	Intranet	Extranet
Definition	Private network accessible only within an organization	Private network that allows limited external access
Users	Employees / internal members	Employees + authorized external users
Access Level	Fully internal	Controlled external access
Purpose	Internal communication & resources	Collaboration with partners
Security	High (restricted access)	High but with external gateways
Network Type	Private LANs & WANs	Private network extended externally
Examples	Company HR system, internal email	Supplier portal, client access system
Access Method	Company login only	Login + partner authorization

1.3 Internet Connections

Internet Access Technologies

Access Type	Description	Typical Users	Key Advantage	Example
Broadband Cable	Uses cable TV infrastructure to provide internet	Home users, small offices	High-speed, easy setup	Home cable modem
DSL (Digital Subscriber Line)	Uses telephone lines for internet	Home users, SOHO	Always-on connection, affordable	ADSL / VDSL
Wireless WAN / Mobile	Internet via cellular networks or wireless signals	Mobile users, remote areas	Mobility, no cables needed	4G / 5G, LTE
Leased Lines / Business DSL	Dedicated line from ISP for organizations	Medium/Large businesses	Reliable, stable, high-speed	Office branch connections
Metro Ethernet	Ethernet service over city networks	Enterprises	High-speed, scalable	Corporate LAN-to-LAN connection

Home & Small Office Internet Connections

Connection Type	Description	Key Advantage	Example
Cable	High bandwidth, always-on, provided by cable TV companies	Fast, stable	Home cable modem
DSL	High bandwidth, always-on, runs over telephone lines	Affordable, always-on	ADSL, VDSL
Cellular	Uses mobile network to connect	Mobile, no wires	4G, 5G internet
Satellite	Provides internet to rural/remote areas	Works where wired ISPs not available	Rural satellite internet
Dial-up	Uses telephone line with modem	Cheap, basic access	Old-school internet

Business Internet Connections

Connection Type	Description	Key Advantage	Example
Dedicated Leased Line	Reserved circuit connecting distant offices	Private, reliable, stable	Corporate office interconnect
Ethernet WAN	Extends LAN technology into WAN	High-speed, scalable	Multi-branch company network
Business DSL	Business-grade DSL, can include symmetric DSL (SDSL)	Always-on, moderate-high speed	Small office DSL
Satellite	Provides connection when wired solution unavailable	Useful for remote branches	Remote site connectivity

Network Types – Traditional and Converged

Feature	Traditional Separate Networks	Converged Network 融合网络
Definition	Different networks for data, voice, video	Single network carrying data, voice, and video
Infrastructure	Separate cabling & equipment for each service	One unified network infrastructure
Technologies	Each service uses different technology	Same technology for all services
Standards / Rules	Each network has its own rules	Single set of rules/standards
Advantages	Dedicated resources per service	Simplifies management, reduces costs
Disadvantages	Expensive, complex, hard to maintain	Requires careful design to handle multiple services
Example	Office has separate LAN, phone network, video network	Modern office uses VoIP, video conferencing, and data on one network

1.4 Reliable Networks

Network Architecture

Feature	Description	Key Points / Examples
Definition	Technologies supporting the infrastructure that moves data	Ensures network works reliably and efficiently
Fault Tolerance	Limits impact of failures by reducing affected devices	- Multiple paths for redundancy - Uses packet-switched networks (packets can take different routes) - Circuit-switched networks do not allow this
Scalability	Ability to expand quickly without affecting existing users	- Supports new users & apps - Follows standards & protocols - Example: Adding new office computers easily
Quality of Service (QoS)	Ensures reliable delivery of critical services	- Manages voice & video traffic - Prevents breaks/pauses in live streaming - Router uses QoS policies to prioritize traffic
Network Security	Protects network infrastructure & transmitted data	Two types: 1. Infrastructure & physical security – prevent unauthorized access 2. Information security – protect transmitted data Three goals: - Confidentiality: only intended recipients can read data - Integrity: data not altered during transmission - Availability: authorized users have reliable access

1.5 Networks Trends

New Networking Trends

Trend	Description	Key Features / Examples
Bring Your Own Device (BYOD)	Users use personal devices for work/school	Devices: laptops, tablets, smartphones, netbooks, e-readers; flexible access anywhere
Online Collaboration	Work with others over network on joint projects	Tools: Cisco WebEx, WebEx Teams; features: instant messaging, post images/videos/links; priority in business & education
Video Communication	Video calls & conferencing for collaboration	Tools: Cisco TelePresence; supports remote communication, critical for teamwork
Cloud Computing	Store/access data & applications over the Internet	Accessible anywhere, enables businesses to deliver to any device; relies on data centres; small companies can lease servers

Cloud Types

Cloud Type	Description	Example / Usage
Public Cloud	Available to everyone via pay-per-use or free	Google Drive, Microsoft OneDrive
Private Cloud	Restricted to specific organization	Government or enterprise private cloud
Hybrid Cloud	Combines two or more cloud types, connected but distinct	Part private, part public cloud for businesses
Custom Cloud	Built for specific industries	Healthcare cloud, media cloud; can be private or public

Home & Small Office Technology Trends

Technology	Description	Key Features / Example
Smart Home	Integrates appliances with network	Ovens linked to calendar, interconnect devices in all rooms
Powerline Networking	Connect devices via electrical outlets	Useful when Wi-Fi or LAN cables can't reach devices; uses standard adapters & specific frequencies
Wireless Broadband	Connect home/business to internet wirelessly	Uses cellular technology (like smartphones); WISP provides hotspots; antenna outside house provides wired/wireless connectivity

Network Management Issues

Issue Area	Key Problems	Explanation / Examples
Network Performance	Bandwidth Congestion 带宽拥塞	Too much traffic slows network
	High Latency 高延迟	Delays affect real-time apps (e.g., VoIP, video)
	Packet Loss 丢包	Data packets not reaching destination → corruption/disruption
	Jitter 抖动	Variable packet arrival → affects voice/video quality
Network Security	Unauthorized Access	Protect network from unauthorized users
	Malware / Viruses	Malicious software protection
	Data Breaches 数据泄露	Prevent sensitive data leaks
	Denial of Service (DoS) Attacks	Ensure network resilience against attacks
Network Reliability	Hardware Failures	Router, switch, cable failures
	Redundancy / Failover 冗余/故障转移	Backup systems ensure continuous operation
	Network Downtime	Minimize planned/unplanned outages
Configuration Management	Configuration Errors	Avoid misconfigured devices causing problems
	Documentation	Keep accurate records of configurations & changes
	Change Management	Plan/document network changes to prevent disruptions
Network Scalability	Capacity Planning	Plan for future growth & traffic increases
	Upgrading Infrastructure	Update hardware/software for new demands
	Load Balancing	Distribute traffic efficiently to avoid overload

CHAPTER 2: Protocols and Models

2.1 Rules of Communications

Communications Fundamentals

Element	Description	Key Idea	Example
Source (Sender)	Device that sends the message/data	Origin of communication	PC sending email
Destination (Receiver)	Device that receives the message	End point of communication	Mail server receiving email
Channel (Media)	Path used to transmit the message	Communication medium	Cable, Wi-Fi, Fiber
Communication Flow: Sender → Channel/Media → Receiver (Example: Laptop → Wi-Fi → Router → Internet → Server)			

Communication Protocols

Term	Description	Key Function	Example
Protocols	Rules governing communication between devices	Define how data is sent, received, and interpreted	TCP, IP, HTTP
Purpose	Ensure devices understand each other	Standardize communication process	Web browsing rules
Variation	Different protocols have different rules	Depends on communication type	Email vs Web protocols

Rule Establishment

Concept	Description	Key Idea	Example
Rule Establishment	Communication must follow agreed rules	Ensures messages are understood	Language formatting rules
Purpose	Makes conversations clear and organized	Avoids confusion/misinterpretation	Proper email format
Formatting Importance	Poor formatting makes messages hard to read	Structured format improves clarity	Paragraphs, punctuation

Protocol Requirements

Requirement	Description	Why Important	Example
Identified Sender & Receiver	Must know who sends and who receives	Ensures correct delivery	Email address (To/From)
Common Language & Grammar	Both sides must understand the format/language	Prevents misunderstanding	English protocol syntax
Speed & Timing	Defines how fast data is sent & synchronized	Prevents delays/collisions	Streaming/video calls
Confirmation / Acknowledgment	Receiver confirms message received	Ensures reliable delivery	TCP acknowledgment reply

Network Protocol Requirements

Requirement		Description	Purpose
Message Encoding		Convert data into signals	Enable transmission over media
Message Formatting & Encapsulation		Structure data into frames	Ensure proper delivery
Message Size		Break data into smaller pieces	Easier transmission
Message Timing		Control speed & delivery timing	Prevent congestion/collision
Message Delivery Options		Define how data is sent	Target correct receivers
Message Encoding			
Aspect	Explanation		Example
Definition	Converting information into transmittable form		Digital data → signals
Signal Types	Depends on media used		Electrical (copper), Light (fiber), Radio waves (wireless)
Decoding	Receiver converts signals back to data		Server reads transmitted message
Message Formatting & Encapsulation			
Aspect	Explanation		Example
Formatting	Message must follow specific structure		Email format vs Web data
Encapsulation 封装	Wrapping data with headers/trailers		Data packaged into frames
Purpose	Ensures proper routing & interpretation		Address info added
Message Size			
Aspect	Explanation		Key Idea
Data Segmentation	Large messages split into smaller parts		Easier network transfer
Frame Transmission	Each piece sent as separate frame		Independent delivery 独立交付
Addressing	Each frame has destination info		Ensures correct routing
Reassembly	Receiver rebuilds original message		Complete data recovery
Message Timing			
Timing Factor	Description	Why Important	Example
Flow Control	Controls transmission rate	Prevents receiver overload	Streaming buffer control
Response Timeout	Waiting time for reply	Detects transmission failure	Retransmit if no reply
Access Method	Determines send timing	Avoids collisions	Ethernet send rules
Collision Handling	Manages simultaneous transmissions	Prevents data corruption	Two PCs send at same time
Message Delivery Options			
Delivery Type	Description	Communication Type	Example
Unicast	One sender → one receiver	One-to-one	Email, web request
Multicast	One sender → selected group	One-to-many (specific)	Video conference group
Broadcast	One sender → all devices	One-to-all	ARP in IPv4
Anycast*	One sender → nearest receiver	One-to-nearest	IPv6 routing

2.2 Network Protocols and Standard

Network Protocol Overview

Protocol Type	Description	Example Use
Network Communications	Allows devices (computers, phones, servers) to communicate over one or more networks. It defines how data is sent and received.	Web browsing, email sending
Network Security	Protects data during transmission by providing authentication, integrity, and encryption.	HTTPS, VPN security
Routing	Helps routers exchange route information, compare paths, and choose the best path to deliver data.	Internet data routing
Service Discovery	Automatically detects devices or services available on a network.	Finding printers, shared folders

Protocol Functions & Interaction

Protocol Functions		
Function	Description	
Addressing	Identifies the sender and receiver (like writing addresses on a parcel).	
Reliability	Ensures data arrives completely and correctly.	
Flow Control	Controls the speed of data transmission so the receiver is not overloaded.	
Sequencing	Labels each data segment so it can be reassembled in order.	
Error Detection	Checks whether data is corrupted during transmission.	
Application Interface	Allows communication between network applications (software-to-software).	
Protocol Interaction (How Protocols Work Together)		
Protocol	Main Function	Simple Explanation
HTTP (Hypertext Transfer Protocol)	Governs web server–client interaction	Defines how web content is requested and delivered.
TCP (Transmission Control Protocol) 传输控制协议	Manages conversations & reliability	Ensures data delivery, sequencing, and flow control.
IP (Internet Protocol)	Global message delivery	Sends data from sender to receiver across networks.
Ethernet 以太网	Local network delivery	Transfers data between NICs within the same LAN.

2.3 Protocol Suites

Network Protocol Suites

Protocol Suite Overview			
Term	Explanation (Easy to Understand)		
Protocol Suite	A group of interrelated protocol (一组相互关联的协议) that work together to perform communication functions.		
Purpose	Provides a complete set of rules to ensure devices can communicate effectively.		
Layered View	Protocols are organized into layers to simplify design and operation.		
Higher Layers	Focus on applications, data representation, and services to users.		
Lower Layers	Concerned with moving data across the network and delivering it physically.		
Explanation: A protocol suite is like a team of protocols , each handling a different job but working together to send and receive data.			
Evolution of Protocol Suites			
Protocol Suite	Developed By	Type	Description
TCP/IP (Internet Protocol Suite)	IETF (Internet Engineering Task Force)	Open Standard	The most widely used suite; foundation of the Internet.
OSI Protocols	ISO & ITU	International Standard Model	Conceptual framework for understanding networking. 理解网络的概念框架
AppleTalk	Apple Inc.	Proprietary	Designed for Apple devices (legacy).
Novell NetWare	Novell Inc.	Proprietary	Used in early enterprise networks.
TCP/IP Protocol Suite			
Feature		Explanation	
Internet Standard		Used globally for Internet communications.	
Open Standard		Freely available for any vendor to use. 任何供应商均可免费使用	
Interoperable		Ensures different devices/vendors can communicate.	
Industry Endorsed		Approved by networking standards organizations.	
TCP/IP Layers & Protocol Examples			
TCP/IP Layer	Function		Protocol Examples
Application	Provides services to user applications		HTTP, FTP, DNS
Transport	Manages conversations and reliability		TCP, UDP
Internet	Handles logical addressing & routing		IP
Network Access	Moves data on physical network		Ethernet, WLAN
TCP/IP Communication Process (Encapsulation)			
Step		Process Explanation	
1. Data Creation		Web server prepares the web page data.	
2. Encapsulation		Data is wrapped with protocol headers (TCP → IP → Ethernet).	
3. Transmission		Data travels across the network.	
4. De-encapsulation		Client removes headers layer by layer.	
5. Data Display		Web browser displays the web page.	

2.3.1 Standards Organizations

Open Standards & Internet Standards

Open Standards Overview	
Term	Explanation
Open Standards	Publicly available standards that allow different devices and systems to communicate.
Interoperability	Ensures products from different vendors can work together.
Competition	Encourages multiple companies to develop compatible technologies.
Innovation	Promotes new ideas and technology development.
Standards Organizations	Vendor-neutral and non-profit bodies that create and promote open standards.
Major Internet Standards Organizations	
Organization	Main Role / Function
ISOC (Internet Society)	Promotes open development and evolution of the Internet.
IAB (Internet Architecture Board)	Manages and oversees Internet standards and architecture development.
IETF (Internet Engineering Task Force)	Develops, updates, and maintains Internet and TCP/IP technologies.
IRTF (Internet Research Task Force)	Conducts long-term research on Internet and TCP/IP protocols.
TCP/IP Supporting Organizations	
Organization	Main Responsibilities
ICANN (Internet Corporation for Assigned Names and Numbers)	Coordinates IP address allocation and domain name management.
IANA (Internet Assigned Numbers Authority)	Manages IP addresses, domain names, and protocol identifiers under ICANN.
Electronic & Communication Standards Bodies	
Organization	Key Focus Areas
IEEE (Institute of Electrical and Electronics Engineers)	Networking standards, telecommunications, power, healthcare technologies.
EIA (Electronic Industries Alliance)	Electrical wiring, connectors, equipment racks.
TIA (Telecommunications Industry Association)	Radio systems, cellular towers, VoIP, satellite communications.
ITU-T (International Telecommunication Union – Telecommunication Standardization Sector)	Video compression, IPTV, broadband (e.g., DSL).

2.3.2 Reference Models

Benefits of Using a Layered Model

Benefit	Description
Simplifies Complex Networking Concepts 简化复杂的网络概念	Breaks down complex network operations into smaller, manageable layers that are easier to understand and explain.
Assists in Protocol Design 协助制定方案	Each layer has defined functions and interfaces, making it easier to design and develop protocols.
Fosters Vendor Competition 促进供应商竞争	Products from different vendors can interoperate because they follow the same layered standards.
Isolates Technology Changes 隔离技术变革	Changes in one layer do not affect other layers, allowing technology to evolve without disrupting the whole system.
Provides Common Language 提供共同语言	Creates a universal terminology (通用术语) to describe network functions and capabilities.

OSI Reference Model

Layer	Description	Example
7 – Application	Provides protocols for process-to-process communication	Web browser requesting a webpage, email client sending email
6 – Presentation	Translates and formats data for the application layer	Converts text, images, or video into a standard format
5 – Session	Manages data exchange between applications	Logging into a website and maintaining session info
4 – Transport	Segments, transfers, and reassembles data reliably	TCP ensures that a large file is sent in order without missing parts
3 – Network	Routes data across networks	IP determines how a packet goes from your PC to a server in another country
2 – Data Link	Transfers frames over physical media and detects errors	Switch sends Ethernet frames within a LAN
1 – Physical	Physical transmission of bits	Ethernet cable, Wi-Fi signal, fiber optics

TCP/IP Reference Model

Layer	Description	Example
Application	Prepares data for the user and handles encoding	Web browsing, email, FTP file transfer
Transport	Ensures communication between devices and reliable delivery	TCP guarantees email or video data arrives intact
Internet	Finds the best path through the network	IP routing a packet from your home to a website server
Network Access	Manages physical devices and media	Ethernet, Wi-Fi, NIC hardware handling actual data transfer

2.4 Data Encapsulation

Segmenting & Multiplexing

Concept	Description	Example / Benefit
Segmenting	Breaking large messages into smaller units for transmission	Sending a large file in multiple parts rather than one huge block
Multiplexing	Interleaving multiple streams of segmented data together	Streaming video and audio at the same time over one network link
Benefits of Segmenting	1. Increases speed – network link not tied up by huge data 2. Increases efficiency – only lost segments are retransmitted	Faster file transfer, reduces retransmission overhead

Sequencing

Concept	Description	Example
Sequencing	Numbering each segment so it can be reassembled in order at the destination	TCP numbers each segment of a file so the receiver can rebuild it correctly
Responsibility	TCP protocol ensures sequencing	Downloading a multi-part email attachment in correct order

Data Encapsulation (PDU) and De-encapsulation

Data Encapsulation (PDU)			
Step	PDU Name	Description	Example
1	Data	Original data stream from the application	Text typed in a web browser
2	Segment	Transport layer adds headers (sequence, port)	TCP segment for a file transfer
3	Packet	Network layer adds addressing info	IP packet with source & destination IP
4	Frame	Data link layer adds MAC addresses and control info	Ethernet frame sent on LAN
5	Bits	Physical layer converts frame into electrical/optical/radio signals	0s and 1s transmitted over cable or Wi-Fi
Data De-encapsulation			
Step	Description		Example
1	Bits received		Physical layer receives signals from network
2	Frame		Data link layer removes frame header
3	Packet		Network layer removes IP header
4	Segment		Transport layer removes TCP header
5	Data		Application layer receives original data stream

2.4.1 Data Access

Addressing at Network & Data Link Layers

Layer	Address Type	Purpose	Example
Network Layer	IP Address (Logical Address)	Deliver packet from original source to destination across networks	Sending an email from your PC (192.168.1.100) to a server in another city (203.0.113.10)
Data Link Layer 数据链路层	MAC Address (Physical Address)	Deliver frame between devices on the same network (LAN)	Ethernet frame sent from your PC's NIC to a switch or another PC on the same LAN

IP Address Structure

Part	Description	Example
Network Portion / Prefix 网络部分/前缀	Identifies the network group the device belongs to	192.168.1.0/24 → all devices on this LAN share 192.168.1
Host Portion / Interface ID 主机部分/接口 ID	Identifies the specific device within the network	PC1 → 192.168.1.110, FTP Server → 192.168.1.9
Same Network Rule	If devices share the same network portion, they are on the same LAN	Both PC1 and FTP server have network portion 192.168.1 → same LAN

Data Link Layer Addresses

Concept	Description	Example
MAC Address	Physical address embedded in the NIC, used for local delivery on the same Ethernet link	PC1 → 00:1A:2B:3C:4D:5E sends a frame to PC2 → 00:1A:2B:3C:4D:6F
Source MAC	MAC address of the device sending the frame on the LAN	PC1 NIC address
Destination MAC	MAC address of the device receiving the frame on the same LAN	PC2 NIC address
Key Rule	MAC addresses only work locally; destination must be on the same link	Even if the ultimate destination is a remote server, the frame first goes to the local gateway/router

Network Layer Addresses

Concept	Description	Example
IP Address	Logical address used to identify devices across networks	PC1 → 192.168.1.110, Web Server → 172.16.1.20
Source IP	IP of the sending device	192.168.1.110
Destination IP	IP of the final target device	172.16.1.20
Role	Network layer ensures packets reach the correct remote network, then hands them to the local data link for final delivery	PC1 sends to Web Server → packet routed through routers to reach remote network

Data Link & Network Layer: Remote Networks

Step	Layer	Source Address	Destination Address	Explanation
1 – First Hop	Data Link (Layer 2)	PC1 MAC	Router (Default Gateway) MAC	PC1 sends frame to the router on the same LAN to start the journey.
	Network (Layer 3)	PC1 IP (192.168.1.110)	Web Server IP (172.16.1.20)	The IP of PC1 and the Web Server stays the same for the whole journey.
2 – Second Hop	Data Link	Router exit interface MAC	Next Router MAC	Router forwards frame to the next router along the path.
3 – Last Hop	Data Link	Last Router exit interface MAC	Web Server NIC MAC	Frame reaches the final LAN; delivered to Web Server.
	Network	PC1 IP	Web Server IP	IP addresses do not change – only the frame (MAC addresses) changes hop by hop.

CHAPTER 3: Physical Layer (OSI Model)

3.1 Purpose of the Physical Layer

Physical Layer & Physical Connection

Concept	Description	Example
Physical Connection	Devices must be physically connected to a network before communicating	Plugging your PC into a LAN cable or connecting via Wi-Fi
Network Interface Card (NIC)	Hardware in a device that allows it to connect to the network	A laptop's Ethernet port or Wi-Fi adapter
Wired vs Wireless	Wired = cables (Ethernet, fiber) Wireless = Wi-Fi or cellular	Wired = fast and stable Wireless = flexible, can move around
Physical Layer Function	Sends bits across the network media	Converts a data frame into electrical signals (Ethernet), light pulses (fiber), or radio waves (Wi-Fi)
Encapsulation Role	Accepts frame from Data Link Layer and encodes it as signals	Data from PC → frame → bits → transmitted over cable/Wi-Fi
Reception by Next Device	Next device receives bits, re-encapsulates frame, then forwards or processes	Router receives signals from PC, rebuilds frame, checks where to send next

3.1.1 Physical Layer Characteristics

Physical Layer – Components, Encoding & Signaling

Area	Description	Example
Physical Components	Hardware devices, media, and connectors that transmit bits	NICs, Ethernet cables, fibre optics, Wi-Fi antennas, RJ45 connectors
Encoding	Converts bits into a recognizable format for the next device	Manchester encoding, 4B/5B, 8B/10B → predictable patterns of signals
Signalling	Represents bits ("1" and "0") on the physical medium	Electrical pulses on copper cables, light pulses on fibre, radio waves for Wi-Fi

Physical Layer – Bandwidth and Performance Terms

Term	Description	Example
Bandwidth	Capacity of a medium to carry data (bits per second)	Fiber cable = 1 Gbps → can carry 1,000,000,000 bits per second
Latency 延迟	Time it takes for data to travel from source to destination	Ping from your PC to a website takes 50 ms
Throughput 吞吐量	Actual amount of data transmitted over a time period	Cable says 100 Mbps bandwidth, but only 80 Mbps is achieved due to network traffic
Goodput 良好产量	Usable data delivered over a time period (removes overhead)	Out of 80 Mbps throughput, only 70 Mbps is actual usable data after headers, acknowledgments, etc.

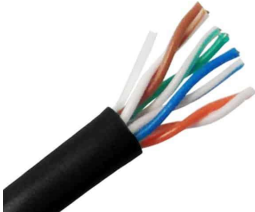
Bandwidth Units

Unit	Abbreviation 缩写	Equivalence 等价
Bits per second	bps	1 bps = 1 bit per second
Kilobits per second	Kbps	1 Kbps = 1,000 bps
Megabits per second	Mbps	1 Mbps = 1,000,000 bps
Gigabits per second	Gbps	1 Gbps = 1,000,000,000 bps
Terabits per second	Tbps	1 Tbps = 1,000,000,000,000 bps

3.2 Network Media

3.2.1 Copper Cabling

Characteristics of Copper Cabling		
Item	Description	Example
Definition	Most common network cable, uses electrical signals to transmit data	Ethernet LAN cable in homes/offices
Advantages	Inexpensive, easy to install, low electrical resistance	Cheap and widely used for PCs and routers
Attenuation	Signal becomes weaker over long distances	Long cable → slower or unstable connection
EMI / RFI	Interference from electromagnetic or radio signals	Nearby power cables or radio devices disrupt signal
Crosstalk	Signals from nearby wires interfere with each other	Two cable pairs "leak" signals
Mitigation Methods	Limit cable length, shielding, twisting wires	Twisted pair design reduces interference

Types of Copper Cabling			
Type	Description	Key Characteristics	Common Use Example
UTP (Unshielded Twisted Pair)	Most common copper cable	• RJ-45 connector • Twisted pairs reduce interference • Color-coded wires • Outer jacket protection	Home/office LAN connections (PC ↔ Switch)
STP (Shielded Twisted Pair)	UTP + extra shielding	• Better EMI/RFI protection • Foil/braided shield • More expensive • Harder to install	Industrial areas, high interference environments
Coaxial Cable	Single copper core with shielding layers	• Inner copper conductor • Plastic insulation • Metallic shield • Outer jacket	Cable internet, TV cable, wireless antennas
UTP (Unshielded Twisted Pair)		STP (Shielded Twisted Pair)	Coaxial Cable
			

3.2.2 UTP Cabling

Properties of UTP Cabling (How UTP Reduces Interference)

Property	Description	Example
Twisted Pairs 双绞线	4 pairs of color-coded copper wires twisted together inside plastic sheath (塑料护套)	Standard LAN Ethernet cable
No Shielding 无屏蔽	UTP does not use metallic shielding	Cheaper and lighter cable
Cancellation	One wire positive, one negative → magnetic fields cancel interference	Like noise cancelling effect
Twist Variation 扭转变换	Each pair twisted at different rates to reduce crosstalk	Prevents signals mixing between pairs

UTP Cabling Standards, Categories, Connectors & Cable Types

Item	Description	Example / Use
Standards Body (TIA/EIA-568)	Defines cable types, length, connectors, termination, testing	Ensures cables work properly
IEEE Standards	Rates performance of copper cables	Cat3, Cat5e, Cat6
Cable Categories	Performance level of UTP cables	Cat5e = Gigabit LAN, Cat6 = Faster LAN
RJ-45 Connector	Plug used to terminate UTP cables	Insert into NIC / Switch port
RJ-45 Socket	Port where RJ-45 plugs in	Router / Switch interface
Cable Types		
Cable Type	Wiring Standard	Application
Straight-Through	Both ends T568A or T568B	Host ↔ Switch / Router
Crossover	One end T568A, other T568B	Host ↔ Host, Switch ↔ Switch
Rollover (Console)	Cisco proprietary wiring	PC ↔ Router/Switch console

The diagram illustrates the wiring for two common UTP standards: T568A and T568B. Each standard shows an 8-pin RJ-45 connector with four color-coded pairs. In T568A, the pairs are connected in order: Pair 1 (blue/white) to pin 1, Pair 2 (orange/white) to pin 2, Pair 3 (green/white) to pin 3, and Pair 4 (brown/white) to pin 4. In T568B, the pairs are crossed: Pair 1 (blue/white) to pin 2, Pair 2 (orange/white) to pin 3, Pair 3 (green/white) to pin 1, and Pair 4 (brown/white) to pin 4.

3.2.3 Fiber-Optic Cabling

Properties & Types of Fiber-Optic Cabling

Category	Description
General Properties	Not as common as UTP due to higher cost
	Ideal for specific networking scenarios
	Transmits data over longer distances at higher bandwidth than other media
	Less susceptible to attenuation
	Completely immune to EMI/RFI
	Made of flexible, extremely thin strands of very pure glass
	Uses laser or LED to encode bits as pulses of light
	Acts as a waveguide to transmit light with minimal signal loss
Single-Mode Fiber (SMF)	Very small core
	Uses expensive lasers
	Designed for long-distance applications
Multimode Fiber (MMF)	Larger core
	Uses less expensive LEDs
	Light travels at different angles
	Supports up to 10 Gbps over 550 meters
Dispersion	Spreading of light pulses over time
	Higher dispersion = greater signal loss
	MMF has more dispersion than SMF
	Maximum MMF distance ≈ 550 meters

Fiber-Optic Connectors

 Straight-Tip (ST) Connectors	 Lucent Connector (LC) Simplex Connectors
 Subscriber Connector (SC) Connectors	 Duplex Multimode LC Connectors

Fiber Patch Cords

 SC-SC MM Patch Cord	 LC-LC SM Patch Cord
 ST-LC MM Patch Cord	 ST-SC SM Patch Cord
A yellow jacket is for single-mode fiber cables and orange (or aqua) for multimode fiber cables.	

Fiber versus Copper

Implementation Issue	UTP Cabling	Fiber-Optic Cabling
Bandwidth Supported	10 Mb/s – 10 Gb/s	10 Mb/s – 100 Gb/s
Distance	Short (1 – 100 m)	Long (1 – 100,000 m)
Immunity to EMI/RFI	Low	High (Completely immune)
Immunity to Electrical Hazards	Low	High (Completely immune)
Media & Connector Cost	Lowest	Highest
Installation Skills Required	Lowest	Highest
Safety Precautions	Lowest	Highest

CHAPTER 4: Data Link Layer (OSI Model)

4.1 Data Link Layer Protocols

4.1.1 Purpose of the Data Link Layer

Purpose and Functions of the Data Link Layer

Function	Description	Simple Example
Communication between NICs	The Data Link Layer allows communication between network interface cards (NICs) on devices	PC NIC sending data to switch NIC
Encapsulation 封装	Encapsulates Layer 3 packets (IPv4/IPv6) into Layer 2 frames	IP packet → Ethernet frame
Access to Physical Media	Allows upper layers to use the physical media (cable, Wi-Fi)	Data sent through Ethernet cable
Error Detection	Detects errors in frames and rejects corrupted frames	Frame check sequence (FCS) detects damaged data
Frame Processing at Each Hop	Routers perform several L2 actions when forwarding data	Router receives frame → removes frame → adds new frame → forwards it

Router Layer 2 Process

1. Accept frame from medium
2. De-encapsulate frame to reveal packet
3. Re-encapsulate packet into a new frame
4. Forward frame to next network segment

Data Link Layer Standards

Data Link Layer Standards	Data Link Layer Standards	Data Link Layer Standards
IEEE	802.2	Logical Link Control (LLC)
	802.3	Ethernet LAN
	802.4	Token Bus
	802.5	Token Ring
	802.11	Wireless LAN (Wi-Fi)
	802.15	Bluetooth (WPAN)
	802.16	WiMAX
ITU-T	G.992	ADSL broadband
	G.8100 – G.8199	MPLS transport
	Q.921	ISDN
	Q.922	Frame Relay
ISO	HDLC	High Level Data Link Control
	ISO 9314	FDDI MAC
ANSI	X3T9.5 / X3T12	Fiber Distributed Data Interface (FDDI)

Data Link Layer Standards (Cont.)

2	Data Link	LLC Sublayer - Logical Link Control - Multiplexing: letting multiple Layer 3 protocols (like IP, ARP, RARP, CLNP, ISIS, etc.) use the same MAC layer.	ETHERNET	IEEE 802.2 standards				
		MAC Sublayer - Media Access Control - MAC addresses (source and destination MAC) - Error detection via FCS - Collision resolution (CSMA/CA, CSMA/CD) - Framing/De-framing - Interaction with physical layer		802.3 Ethernet	802.3u FastEthernet	802.3ab GigabitEthernet	802.3ae 10 GigabitEthernet	802.3ba 40 GigabitEthernet
1	Physical	Sending raw bits over the network medium - Light or radio signals (copper, fiber or wireless) - Electrical signals (voltage levels, timing, etc.) - Modulation methods (how bits are turned into signals) - Pinouts and physical layouts (RJ-45, SFP, etc.)						

4.2 Topologies and Media Access Control

Physical and Logical Topologies

Topology Type	Description	Simple Example
Physical Topology	Shows the real physical connections between devices and cables in a network	PCs connected to a switch using Ethernet cables
Logical Topology	Shows how data actually flows in the network using interfaces and IP addressing	Devices communicating through IP routing even if physically connected differently
• Physical topology -> How devices are physically connected		
• Logical topology -> How data moves in the network		

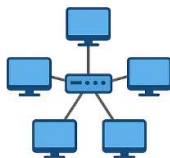
Common WAN and LAN Topologies

Topology	Type	Description	Example
Point-to-Point	WAN	Direct connection between two devices	Branch office ↔ Head office
Hub and Spoke	WAN	Central hub connects multiple branch sites	Headquarters connected to many branch offices
Mesh	WAN	Every device connects to every other device	Internet backbone networks
Star	LAN	All devices connect to a central switch	Office network with one switch
Extended Star	LAN	Multiple star networks connected together	Multiple switches connected to a main switch
Bus	LAN (Legacy)	All devices share a single backbone cable	Early Ethernet networks
Ring	LAN (Legacy)	Each device connected in a circular path	Token Ring networks

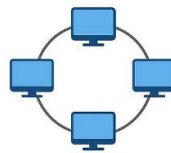
Tips:

- **Point-to-Point** → 2 devices only
- **Hub & Spoke** → 1 centre, many branches
- **Mesh** → everyone connected to everyone
- **Star** → all devices connect to a switch
- **Bus** → one long cable
- **Ring** → circle network

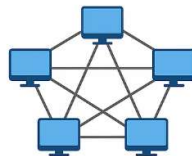
Star Topology



Ring Topology



Mesh Topology



Bus Topology



Half-Duplex vs Full-Duplex Communication

Communication Type	Description	Example
Half-Duplex	Only one device can send or receive data at a time	Walkie-talkie communication
	Devices must wait before sending data	Old Ethernet with hubs
	Often used in shared networks	Wireless LAN (Wi-Fi)
Full-Duplex	Devices can send and receive data at the same time	Phone call conversation
	No collisions because transmission happens simultaneously	Ethernet switches

Tips:

- **Half-duplex** -> one direction at a time
- **Full-duplex** -> both directions at the same time

Access Control Methods and CSMA

Access Method	Full Name	How It Works	Used In	Simple Explanation
CSMA/CD (Duplex Mode: Half-duplex)	Carrier Sense Multiple Access with Collision Detection	Device listens to the network first. If the channel is free, it sends data. If a collision happens, devices stop, wait a random time, then resend.	Legacy Ethernet (IEEE 802.3, with hubs)	Devices share the network, collisions happen, and data is resent.
CSMA/CA (Duplex Mode: Half-duplex)	Carrier Sense Multiple Access with Collision Avoidance	Device checks if the wireless channel is free before sending. It also announces how long it will use the channel so others wait.	Wireless LAN (IEEE 802.11 Wi-Fi)	Devices try to avoid collisions by waiting and reserving the channel.
Controlled Access (Duplex Mode: Usually Half-duplex)	Deterministic Access	Each device has a specific time to use the network, so there's no competition for it.	Token Ring, ARCNET (legacy networks)	Devices take turns sending data instead of competing.

Controlled Access

Feature	Description
Transmission	Only one device can transmit at a time.
Access Method	Devices must wait their turn to transmit, avoiding competition.
Collisions	No collisions occur as devices don't transmit simultaneously.
Access Control	May use token passing to decide which device can transmit next.
Technologies	- Token Ring (IEEE 802.5) - Fiber Distributed Data Interface (FDDI)
Example	Legacy Token Ring LANs are an example of controlled access.
Key Concept	Each device has its own time to use the medium, ensuring no conflicts.

4.3 Data Link Frame

The Frame	
Frame Part	Description
Header	Contains control information (addressing, type, control). Located at the start of the frame.
Data	Contains the actual payload (IP header, transport header, application data).
Trailer	Contains control information for error detection. Located at the end of the frame.
Frame Fields	
Field	Description
Frame Start and Stop	Identifies the beginning and end of the frame.
Addressing	Indicates the source and destination nodes.
Type	Identifies the Layer 3 protocol being encapsulated (e.g., IP).
Control	Identifies flow control services.
Data	Contains the actual payload being transmitted.
Error Detection	Used to determine transmission errors.
Data Link Frame Addresses	
Feature	Description
Physical Address	Used for local delivery of frames on the link. Contained in the frame header.
Forwarding	Updated by each device that forwards the frame.
LAN and WAN Frames	
Protocol	Used For
Ethernet	Common for local area networks (LANs).
802.11 Wireless	Used for wireless networks (Wi-Fi).
PPP	Point-to-point communication.
HDLC	High-Level Data Link Control, used for serial communication.
Frame-Relay	Used for wide area networks (WANs).

CHAPTER 5: Ethernet

5.1 Ethernet Protocol

What is Ethernet

Topic	Meaning	Explanation	Example
Ethernet	Networking technology	Ethernet is used to connect devices in a LAN (Local Area Network). It works at Data Link Layer and Physical Layer .	Connecting laptop to router using LAN cable
IEEE Standards	Rules for Ethernet	Ethernet follows IEEE 802.2 (LLC) and IEEE 802.3 (MAC) standards to ensure all devices communicate correctly	All computers use same format to send data

Data Link Layer Sublayers

Sublayer	Function	Explanation	Example
LLC (Logical Link Control)	Identify protocol	Adds information to tell which Network Layer protocol is used (IP, ARP, etc.)	"This data is IPv4"
MAC (Media Access Control)	Control data transmission	Handles frame creation, MAC addressing, and media access	Sending data from PC to switch
Tips: • LLC -> "What is inside?" • MAC -> "Where to send & how to send?"			

MAC Sublayer

Data Encapsulation			
Function	Meaning	Explanation	Example
Frame	Data structure	Data is wrapped into a frame before sending	Like putting a letter into an envelope
Addressing	MAC Address	Each device has a unique MAC address (like ID)	Laptop → Printer
Error Detection	FCS	Uses CRC (Cyclic Redundancy Check) to detect errors	Detect corrupted file
Media Access Control			
Function	Explanation		Example
Placement of frames	Decides when device can send data		Avoid sending at same time
Removal of frames	Receives data from medium		Device reads incoming data
Media recovery	Handles transmission errors		Retry sending if needed
Tips:			
• Frame -> parcel • MAC address -> home address • FCS -> quality check			

Ethernet Communication Types

Type	Meaning	Explanation	Example
Half-Duplex	One direction only	Devices cannot send and receive at the same time	Walkie-talkie
Full-Duplex	Two directions	Devices send and receive simultaneously	Phone call
Important: • Old networks (Hub) → Half-duplex • Modern networks (Switch) → Full-duplex			

CSMA/CD (Collision Handling)

Term	Meaning	Explanation	Example
CSMA/CD	Collision detection	Device checks if network is free before sending data	Two PCs try to send → collision
Collision	Data crash	Happens when two devices send at same time	Data becomes corrupted
Solution	Wait & resend	Device waits random time and tries again	Retry sending file
Note: • Used in old Ethernet (hub) • Not needed in modern switch networks			

Ethernet Frame Size

Type	Size	Explanation	Example
Minimum Frame	64 bytes	Smaller than this = error (runt frame)	Broken packet
Maximum Frame	1518 bytes	Standard maximum size	Normal Ethernet data
Jumbo Frame	>1500 bytes data	Used in high-speed networks	Large file transfer
Invalid Frame	Too small or too large	Automatically dropped	Network ignores it

Ethernet Frame Fields (Structure of Ethernet Frame)

Field	Function	Detailed Explanation	Example
Preamble	Synchronization	7 bytes pattern to prepare receiver	“Get ready” signal
SFD (Start Frame Delimiter)	Start of frame	Marks the beginning of frame	“Start now”
Destination MAC	Receiver	Address of target device	Printer MAC
Source MAC	Sender	Address of sender	Laptop MAC
Length/Type	Data info	Shows size OR protocol type	IPv4 / 100 bytes
Data	Actual content	Payload (real data)	File, message
FCS	Error detection	Checks if data has errors	CRC check

Example of Ethernet Frame Transmission Process

Scenario: Sending a file from Laptop → Printer	
Step	What happens
1	Laptop creates data (file)
2	Data is wrapped into Ethernet frame
3	Source MAC = Laptop, Destination MAC = Printer
4	Frame sent through cable
5	Printer receives frame
6	Printer checks FCS (error check)
7	If correct → accept data
8	If error → discard frame

5.1.2 Ethernet MAC Address

MAC Address Basics

Topic	Meaning	Explanation	Example
MAC Address	Unique device ID	A 48-bit address assigned to every Ethernet device (NIC, switch, printer)	00:1A:2B:3C:4D:5E
Format	Hexadecimal	12 hex digits (4 bits per digit, 6 bytes total)	00-1A-2B-3C-4D-5E, 001A.2B3C.4D5E
Hexadecimal	Base-16 system	Uses 0–9 and A–F. Easier than binary (4 bits → 1 hex)	0x73 = 01110011 in binary
Purpose	Device identification	Used at the data link layer to tell devices apart	Your laptop sends data to printer NIC

Structure of Ethernet MAC Address

Part	Length	Meaning	Example
OUI (Organizationally Unique Identifier)	3 bytes (24 bits)	Assigned to vendor by IEEE, identifies manufacturer	00:1A:2B
Vendor-assigned	3 bytes (24 bits)	Unique ID assigned by vendor for each device	3C:4D:5E
Full MAC	6 bytes (48 bits)	Combines OUI + vendor ID	00:1A:2B:3C:4D:5E
Explanation: - All devices from the same vendor share the same OUI in the first 3 bytes. - The last 3 bytes are unique per device, ensuring no two devices have the same MAC.			

Frame Processing

Function	Detailed Explanation	Example
Burned-In Address (BIA)	MAC stored in NIC ROM permanently	When laptop starts, NIC copies MAC to RAM
Frame Header	Attached to Ethernet frame	Source MAC = sender, Destination MAC = receiver
Frame Forwarding	NIC checks destination MAC	If match → accept; if no match → discard
Example: - Laptop sends a file to printer: - Source MAC = Laptop NIC - Destination MAC = Printer NIC - Printer NIC compares → accepts frame → data passed up OSI layers		

Types of MAC Addresses

Type	Meaning	Explanation	Example
Unicast	One-to-one	Sent from a single device to a single device	Laptop → Printer
Broadcast	One-to-all	Sent to all devices on LAN	Destination MAC = FF:FF:FF:FF:FF:FF
Multicast	One-to-many	Sent to a group of devices	MAC starts with 01:00:5E → group of devices

5.2 LAN Switches

5.2.1 The MAC Address Table

Ethernet Switch Basics

Topic	Meaning	Detailed Explanation	Example
Ethernet Switch	Layer 2 Device	A device that forwards data frames based on MAC addresses.	Switch connecting 4 PCs in a LAN
MAC Address Table (CAM Table)	Storage of MAC addresses	Switch uses this table to map MAC addresses to specific ports.	Switch stores MAC addresses and associated ports

Learning MAC Addresses (Step-by-Step)

How Switch Learns Source MAC Addresses:

Step	Description	Example
1. Learn Source MAC	Switch examines incoming frames for new source MAC addresses.	Frame from PC1 to Switch: Switch learns MAC address of PC1.
2. Add to Table	If source MAC is unknown, add it to the MAC address table with port number.	PC1's MAC → Port 1
3. Refresh Timer	If source MAC already exists, the switch updates the refresh timer.	PC2's MAC → Port 2 → Table entry updated
4. Default Timeout	Most switches keep the entry for 5 minutes by default.	If no traffic from PC1 after 5 minutes → Entry removed.

Forwarding Frames Based on Destination MAC

Destination MAC	Action	Explanation	Example
Broadcast	Flood the frame to all ports	Broadcast frames go to every device in the network.	ARP request sent to all devices.
Multicast	Flood the frame to all ports	Frames for multicast addresses go to all devices in the multicast group.	Streaming video sent to a group of devices.
Unicast	Look in the MAC address table for the destination	Switch checks the table to find the correct port for the destination MAC.	Frame from PC1 → PC2 → Switch forwards to Port 3.
Unknown Unicast	Flood the frame to all ports	If the destination MAC is unknown, the switch floods the frame out all ports except the incoming port.	Frame from PC3 → Destination MAC unknown → Flood to all ports except Port 2.

MAC Address Table in Action (MAC 地址表实际应用)

Function	Action	Example
Learning Source MAC	Switch learns the source MAC address and updates the table.	Frame from PC4 → MAC address added to table with Port 4.
Forwarding Frames	Switch forwards the frame based on destination MAC address.	Frame with destination MAC of PC1 → Switch forwards to Port 1.
Filtering Frames	Switch filters the frame if the destination MAC address is in the table.	PC1 sends data to PC4 → Switch forwards frame only to Port 4.
Flooding	If destination MAC address is not in the table (unknown unicast), flood the frame to all ports except the incoming one.	Frame for unknown MAC address → Flood to all ports except the incoming port.

5.2.2 Switch Speeds and Forwarding Methods

Switch Forwarding Methods

Method	How It Works	Key Points	Example
Cut-Through Switching	Switch reads only the destination MAC address and forwards immediately.	- Lowest latency- No error checking- Forwarding begins before whole frame is received	PC1 → PC2: Switch forwards as soon as it reads destination MAC
Fast-Forward (variant)	Immediately forwards packet after reading destination MAC	Fastest form of cut-through switching	Used in high-speed LANs for minimal delay
Fragment-Free (variant)	Switch stores first 64 bytes of frame before forwarding	Avoids forwarding frames with collisions in header	Compromise between speed and reliability
Store-and-Forward	Switch receives entire frame, checks for errors (FCS), then forwards	Ensures corrupted frames are not forwarded	Used in reliable enterprise networks
Explanation: • Cut-through → speed over reliability • Store-and-forward → reliability over speed			

Memory Buffering Methods on Switches

Buffering Method	How It Works	Key Points	Example
Port-Based Memory	Each port has its own queue for incoming/outgoing frames	- Frame waits until previous frames in queue are transmitted- Simple but less efficient under congestion	Port 1 queue: PC1 sends 5 frames → transmitted in order
Shared Memory	All ports share a common buffer for frames	- More efficient under heavy traffic- Memory dynamically allocated to ports	Shared buffer stores frames from PC1, PC2, PC3 → forwarded as port becomes available
Explanation: • Port-based → each port isolated • Shared memory → all ports share resources → better for busy networks			

Duplex and Speed Settings

Setting	Meaning	Key Points	Example
Full-Duplex	Send and receive simultaneously	Highest performance	Switch ↔ PC → both send/receive at same time
Half-Duplex	Send or receive at a time	Older hubs or legacy devices	Switch ↔ Hub → only one direction at a time
Autonegotiation	Devices automatically select speed & duplex	Ensures best compatible performance	Switch port detects PC supports full-duplex → auto-configures
Duplex Mismatch	One end full-duplex, other half-duplex	Causes collisions, slow performance	Switch full-duplex ↔ PC half-duplex → performance issues
Notes: • Gigabit Ethernet always full-duplex • Avoid duplex mismatch to maintain high performance			

Auto-MDIX (Automatic Cable Detection)

Feature	How It Works	Key Points	Example
Auto-MDIX	Switch detects type of cable (straight-through or crossover) and configures interface automatically	- Eliminates need for specific cable type- Enabled by default on modern Cisco switches	PC ↔ Switch: straight cable works even if crossover needed
Manual Cross-Over	Before auto-MDIX, required for direct router-to-host or switch-to-switch connections	Old method, manual setup needed	Router ↔ Host → cross-over cable required

5.3 Address Resolution Protocol

5.3.1 MAC and IP

Primary Addresses on an Ethernet Device

Address Type	Layer	Purpose	Example
MAC Address	Layer 2 (Data Link)	Used for NIC-to-NIC communication on the same network	00:1A:2B:3C:4D:5E
IP Address	Layer 3 (Network)	Used to send packets from source device to destination device , even across networks	192.168.1.10
Explanation: • MAC = "hardware address" → identifies physical NIC on LAN • IP = "logical address" → identifies device on network or across networks			

Destination on Same Network

Scenario	Destination MAC	Destination IP	How It Works
Same LAN / Subnet	MAC address of destination device	IP of destination device	Frame sent directly from source NIC to destination NIC using MAC
Example: PC-A (192.168.1.10) → PC-B (192.168.1.20) on same LAN → destination MAC = PC-B's MAC			
Steps: 1. PC-A wants to send data to PC-B on same subnet. 2. PC-A uses ARP (IPv4) or ICMPv6 (IPv6) to find PC-B's MAC address. 3. Frame is delivered directly to PC-B using its MAC address.			

Destination on Remote Network

Scenario	Destination MAC	Destination IP	How It Works
Remote network	MAC address of default gateway	IP of destination device	Frame sent to router (gateway). Router forwards packet to destination network.
Example: PC-A (192.168.1.10) → Web Server (10.0.0.50) → destination MAC = Router R1 interface connected to PC-A's network			
Steps: 1. PC-A wants to send a packet to a device outside its subnet. 2. PC-A knows the IP of the destination (e.g., 10.0.0.50) but not the MAC. 3. PC-A sets destination MAC = default gateway's MAC (Router interface). 4. Router forwards the packet toward the remote network.			

Address Resolution Mechanisms

Mechanism	Protocol	Purpose	Example
ARP (Address Resolution Protocol)	IPv4	Finds MAC address corresponding to IPv4 address	PC-A wants 192.168.1.20 → ARP → finds MAC 00:1A:2B:3C:4D:5E
ICMPv6 / Neighbor Discovery	IPv6	Finds MAC address corresponding to IPv6 address	PC-A wants fe80::1 → ND → finds MAC 00:1A:2B:3C:4D:5F
Note: - Without ARP/ICMPv6, device cannot send frame on Ethernet. - This applies to both same network and remote network (gateway MAC used for remote).			

5.3.2 ARP

ARP Introduction (ARP 介绍)

Concept	Explanation	Example
ARP (Address Resolution Protocol)	ARP is used to map IPv4 addresses to MAC addresses .	PC-A wants to send data to PC-B using IP address, but needs to know PC-B's MAC address first.
ARP Table / Cache	Stores mappings of IPv4 addresses to MAC addresses. This table is stored in RAM on devices.	PC-A stores an entry for PC-B: 192.168.1.20 → 00:1A:2B:3C:4D:5E.
Function	ARP helps devices on the same local network find each other's MAC address by sending ARP requests .	PC-A broadcasts an ARP request for PC-B's MAC address.

ARP Request and Reply Process

ARP Request Process		
Step	Description	Example
1. ARP Request	Device needs to find MAC address for a known IPv4 address. If not in ARP table, it sends a broadcast ARP request.	PC-A needs to find the MAC address for 192.168.1.20, so it broadcasts an ARP request.
2. Broadcast	ARP request is broadcast to all devices in the local network.	The request is sent to every device: "Who has IP 192.168.1.20?"
3. ARP Reply	Only the device with the matching IP address will respond with its MAC address.	PC-B with 192.168.1.20 sends back a reply: "I have 192.168.1.20, my MAC is 00:1A:2B:3C:4D:5E".
ARP Reply Process		
Step	Description	Example
1. ARP Reply	The device that has the requested IP address sends an ARP reply with its MAC address.	PC-B sends an ARP reply to PC-A: 192.168.1.20 → 00:1A:2B:3C:4D:5E.
2. Update ARP Table	The sender updates its ARP table with the received MAC address and associates it with the destination IP.	PC-A now has: 192.168.1.20 → 00:1A:2B:3C:4D:5E in its ARP table.

ARP Functions and Role in Communication

Function	Explanation	Example
Resolve IP to MAC	ARP resolves an IPv4 address to the corresponding MAC address of the destination device.	PC-A resolves 192.168.1.20 to MAC address 00:1A:2B:3C:4D:5E.
Local Network Communication	If destination device is on the same network, ARP resolves its MAC address directly.	PC-A sends a frame directly to PC-B using 192.168.1.20 → MAC-00:1A:2B:3C:4D:5E.
Remote Communication	If destination device is not on the same network, ARP resolves the MAC address of the default gateway (router).	PC-A sends data to a device on another subnet, resolves gateway's MAC, and sends frame to router.

ARP Table Management (ARP 表管理)

Concept	Explanation	Example
Time-based Entries	ARP table entries are not permanent and have an expiration time (default 5 minutes).	After 5 minutes, PC-A's ARP entry for 192.168.1.20 may expire.
Manual Removal	ARP table entries can be manually removed by the network administrator.	arp -d 192.168.1.20 command on Windows to delete entry for 192.168.1.20.
ARP Table Display Commands • Cisco Router: show ip arp • Windows PC: arp -a ▪ Cisco: 192.168.1.1 → a0e0.af0d.e140 ▪ Windows: 192.168.1.1 → c8-d7-19-cc-a0-86		

ARP Issues

ARP Broadcasts and Impact		
Issue	Explanation	Example
Excessive ARP Broadcasts	If many devices send ARP requests at once, the network can become congested with broadcast traffic.	In a large office, every new device connects and sends ARP requests, flooding the network.
ARP Flooding	A large number of ARP requests can reduce the performance of the local network.	Thousands of ARP requests flood the network during a startup.
ARP Spoofing / Poisoning		
Issue	Explanation	Example
ARP Spoofing	Malicious attacker sends fake ARP replies to associate their MAC address with the IP address of a legitimate device, like the gateway.	Attacker (Host C) replies to ARP request for the gateway's IP, causing PC-A to send traffic to Host C instead of the gateway.
ARP Poisoning	An attacker "poisons" the ARP cache, causing traffic to be misdirected to malicious devices.	Host C intercepts traffic intended for the gateway by pretending to be the gateway through ARP poisoning.
Mitigation of ARP Issues		
Mitigation Method	Explanation	Example
Dynamic ARP Inspection (DAI)	Switches inspect ARP packets to prevent spoofing attacks.	Cisco switches configured with DAI filter out ARP spoofing attempts.

ARP in Remote Communication

Scenario	Explanation	Example
Same Network	If the destination is on the same network, the source device sends the frame directly to the destination using the destination's MAC address.	PC-A to PC-B on the same subnet uses MAC address of PC-B.
Remote Network	If the destination is on a remote network, ARP is used to resolve the gateway's MAC address. The source device sends data to the gateway first, and the gateway forwards it to the remote destination.	PC-A → Gateway R1 → Web Server, resolves R1's MAC address using ARP.

ARP Process (Summary)

Step	Action	Details
1. ARP Request	Device needs the MAC address of a destination. If not in the ARP table, it broadcasts an ARP request.	"Who has 192.168.1.20?"
2. ARP Reply	The device with the matching IP responds with its MAC address.	"I have 192.168.1.20, MAC: 00:1A:2B:3C:4D:5E."
3. MAC Address Update	The sender updates its ARP table with the received MAC address.	The sender now associates 192.168.1.20 with 00:1A:2B:3C:4D:5E.
4. Frame Sent	The sender uses the resolved MAC address to forward the data frame.	PC-A sends data frame to PC-B using the MAC address of PC-B.

